UWB WEBINAR SERIES

RESTRICTED

SESSION 2:
Introducing Ultra-Wideband (UWB) Radar

CAS HAMBURG
JULY 2022

SECURE CONNECTIONS
FOR A SMARTER WORLD

Provided under NDA only
COMPANY PROPRIETARY
Hella KGaA Hueck & Co.
a6c1ac-6480-49a3-b722-4

CONTENT OVERVIEW

- Radar
- UWB Radar Pulse
- IEEE Frame Format
- UWB Radar Receiver Concept
- Signal Processing
- Q&A

under NDA only
OMPANY PROPRIETARY
Hella KGaA Hueck & Co
a0fc1ac-6480-4933-b722-4b9e92ea54ae

Radar

SECURE CONNECTIONS
FOR A SMARTER WORLD

Provided under NDA only
COMPANY PROPRIETARY
Hella GaA Hueck & Co.
afc1ac-6480-49a3-b722-4b9e92ea34ae

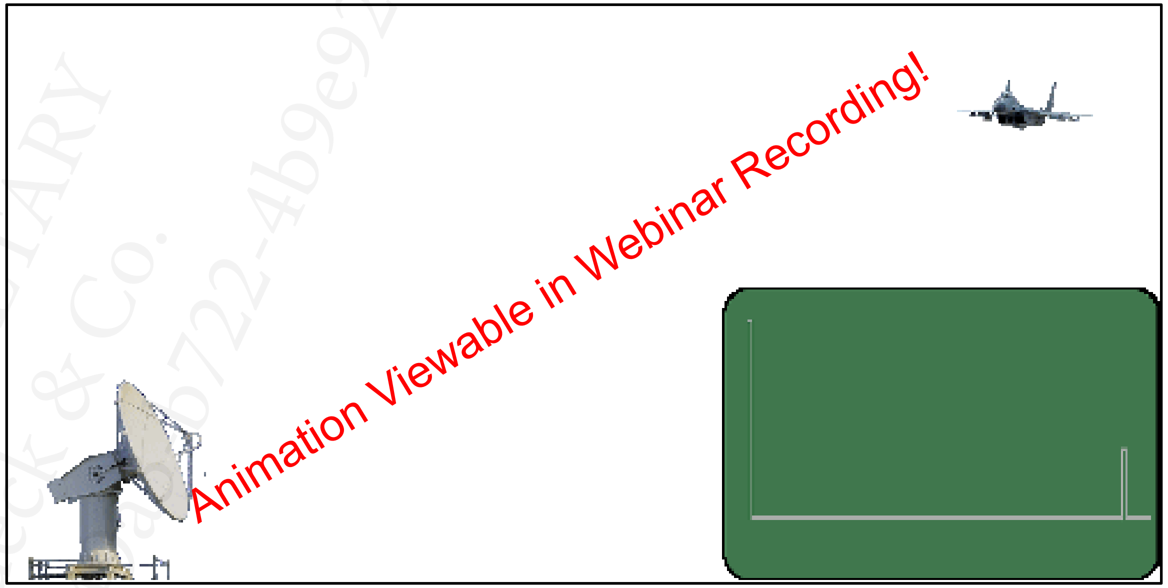
SIMPLE PULSED RADAR

- RADAR (radio detection and ranging)
- Time between Radar and Target:

  ΔT = Round trip time
- Speed of the transmitted signal:

  c = Speed of light
- Range between Radar and Target:

  $$R = \frac{c * \Delta T}{2}$$


Animation Viewable in Webinar Recording!

Animation source: https://www.radartutorial.eu/

PULSED DOPPLER RADAR

- Transmitting multiple pulses enables the doppler effect, allowing to resolve the velocity of a target
- Frequency of signal transmission:

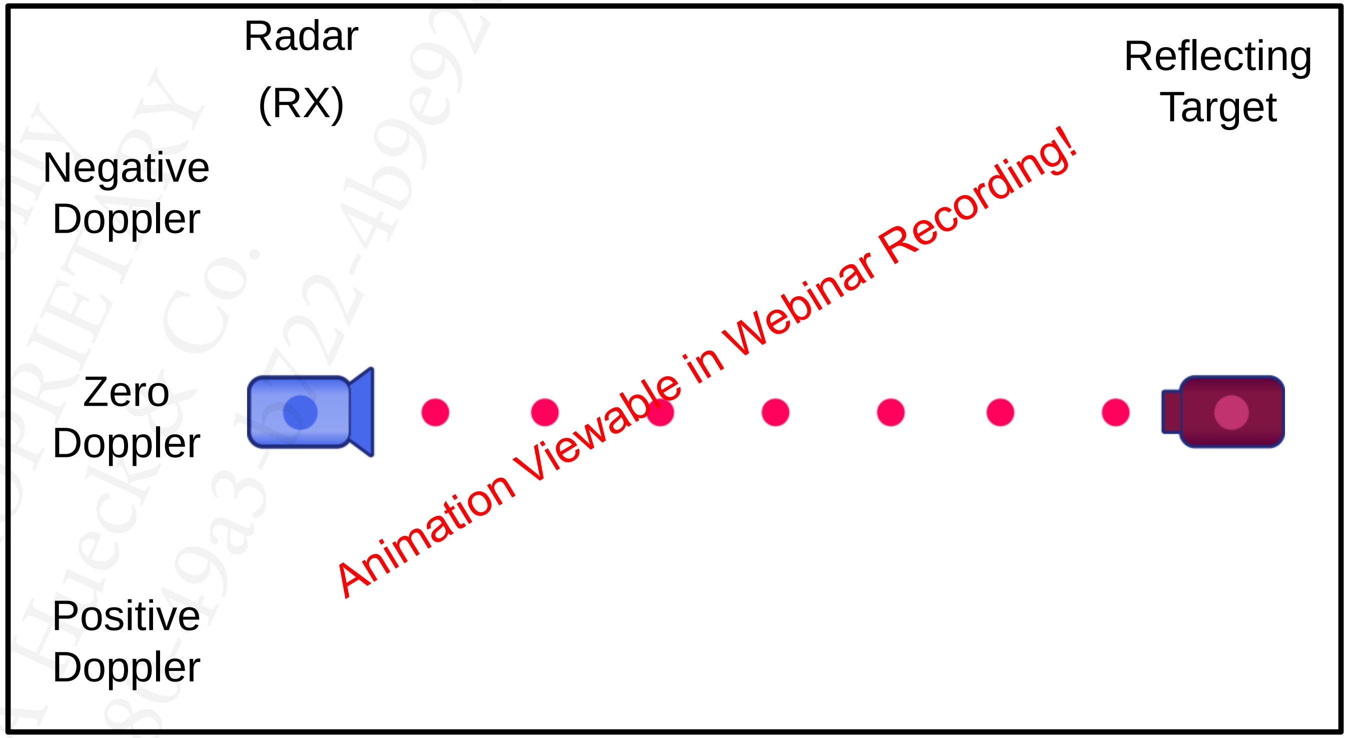
  f_t = Transmission frequency

- Difference between transmission frequency and shifted frequency due to a moving target:

  $f_D = f_{shift} - f_t$ = Doppler frequency

- Velocity of target relative to the Radar:

  $v = \frac{c * f_D}{2 f_t}$


Radar
(RX)
Reflecting Target
Negative Doppler
Zero Doppler
Positive Doppler

Animation Viewable in Webinar Recording!

Animation source: https://www.einstein-online.info/

Provided under NDA only
COMPANY PROPRIETARY
Hella KGaA Hueck & Co.
a0fc1ac-6480-49a3-a722-4b9e92ea34ae

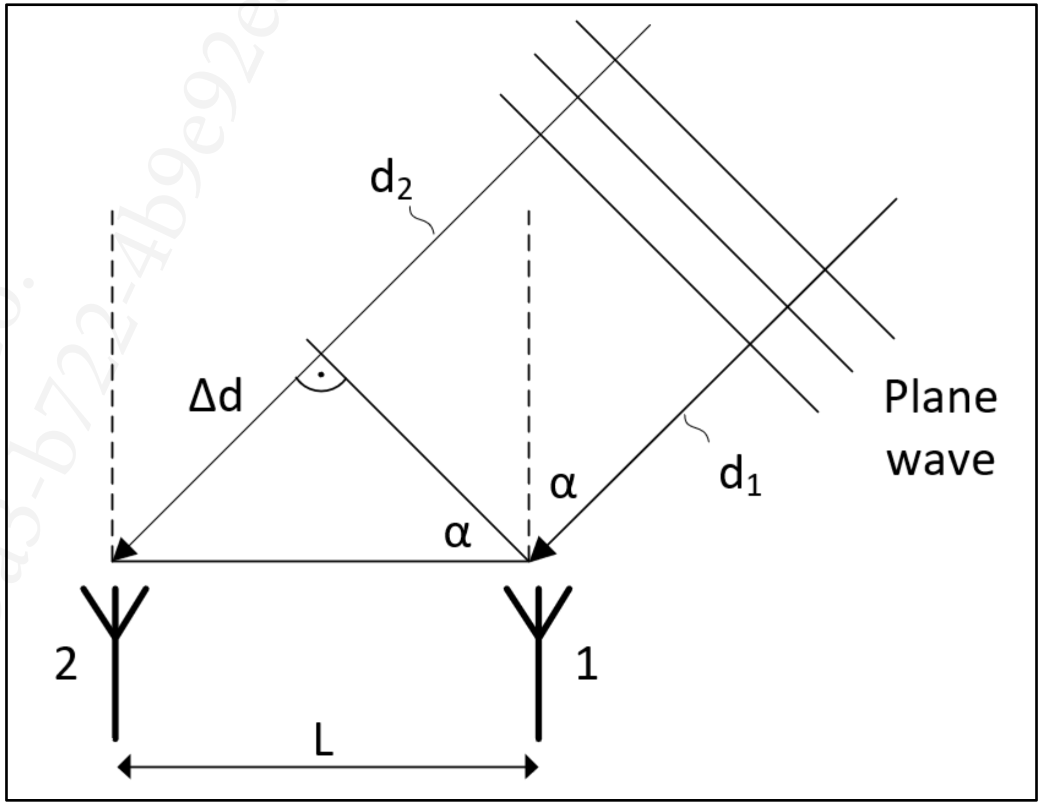
MULTI-CHANNEL RADAR

- Receiving from multiple antennas located in space allows to resolve the Angle of Arrival (AoA) of a target due to the increased path distance between the Radar antennas and the target
- The length between the center of the two antennas:

  L = Distance between the antennas
- Additional distance to get to the second antenna

  $\Delta d = \mathrm{c} * \Delta\tau$ = speed of light * additional trip time
- The Angle of Arrival of the target relative to the Radar at boresight

  $\alpha = \arcsin(\frac{c*\Delta\tau}{L})$



UWB Radar Pulse

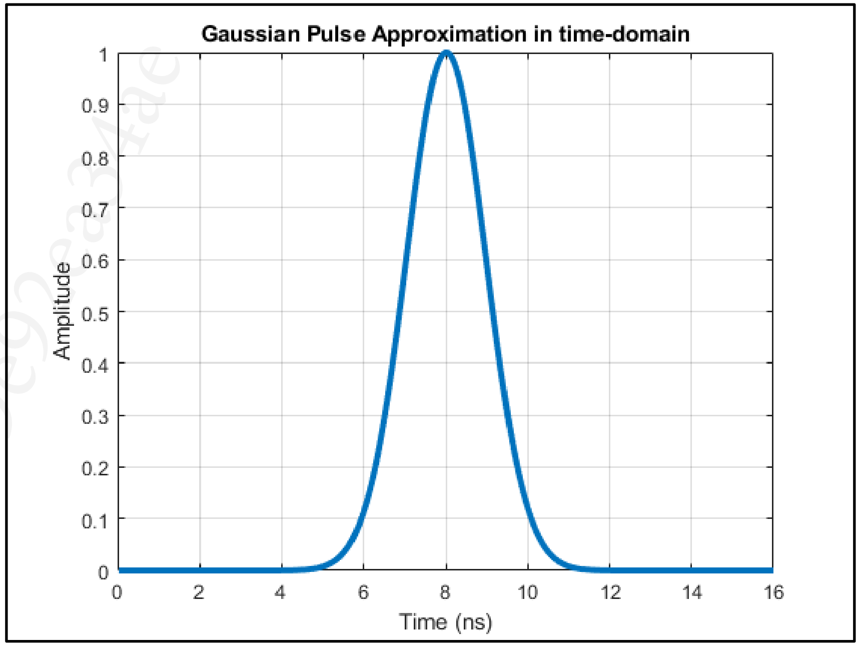
SECURE CONNECTIONS
FOR A SMARTER WORLD

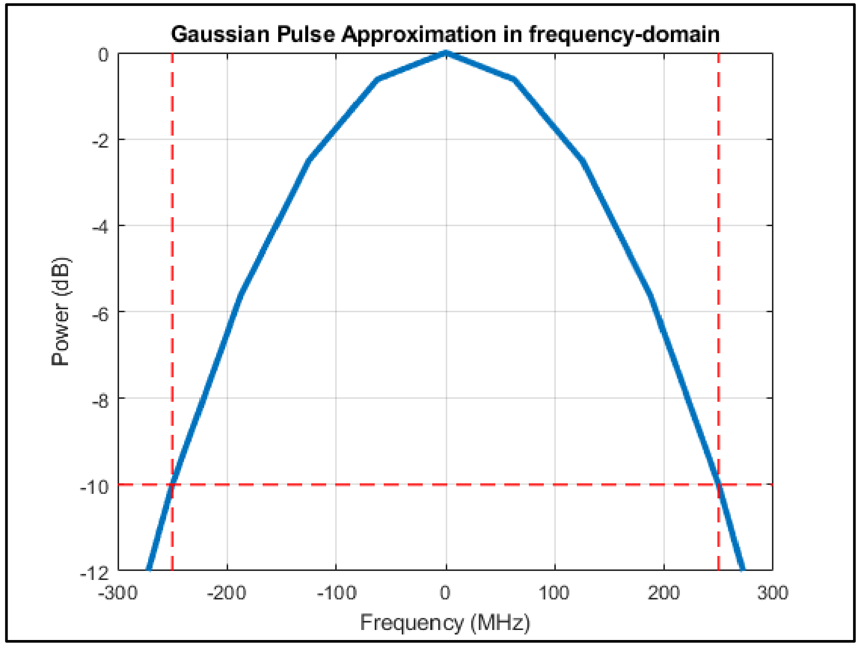
Provided under NDA only
COMPANY PROPRIETARY
Hella GmbH & Co.
afc1ac-6480-49a3-b722-4b9e92ea34ae

UWB RADAR PULSE SHAPE

- Optimal Pulse Shape that estimates a Gaussian Window
- No post- or pre-ringing -> reduces artefacts in measurements
- Remains compliant with IEEE reference pulse
- Bandwidth of the UWB Radar Pulse:

  B= ~ 500 MHz



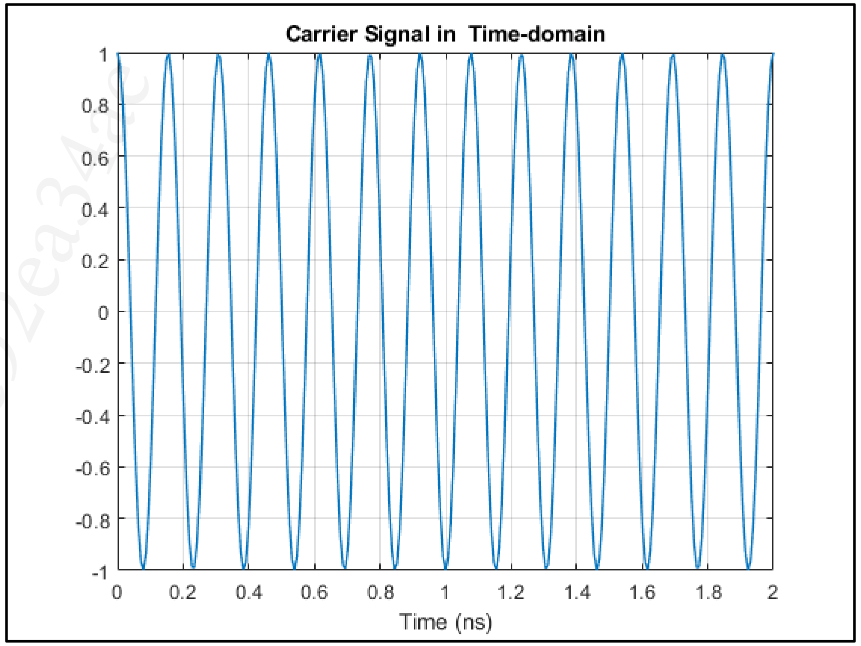


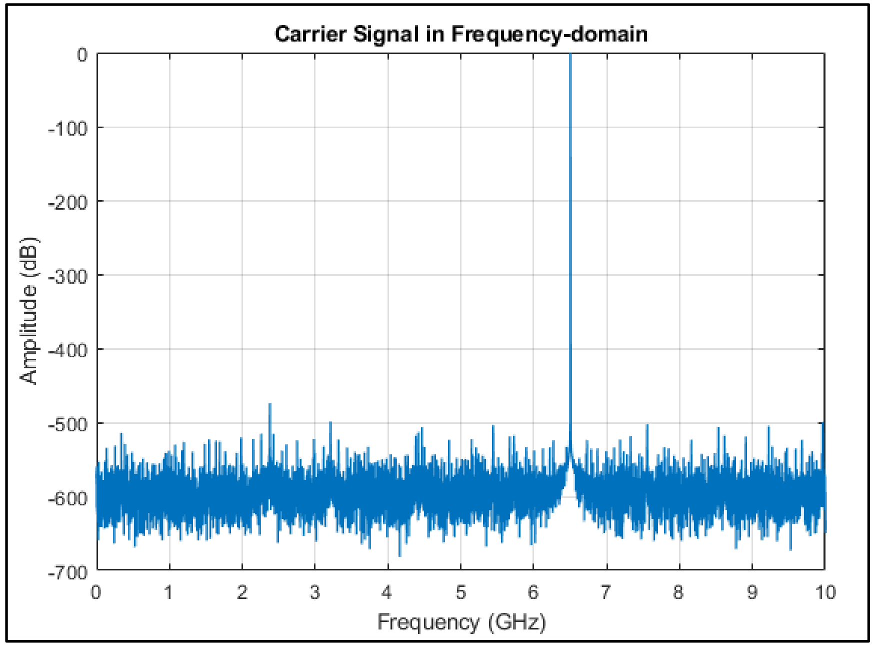
Provided under NDA only
COMPANY PROPRIETARY
Hella KGaA Hueck & Co.
a0fc1ac-6480-49a3-b722-4b9e92ea3fae

CARRIER SIGNAL

- Carrier frequency defines the channel:
  5: 6489.6 MHz
  6: 6988.8 MHz

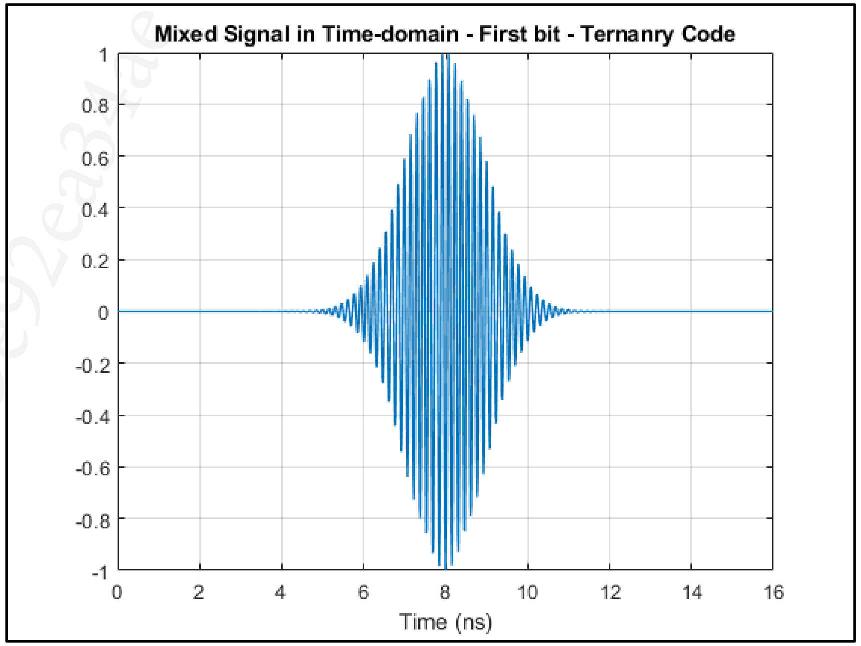
  8: 7488.0 MHz
  9: 7987.2 MHz
- Is a simple cosine wave with a constant frequency

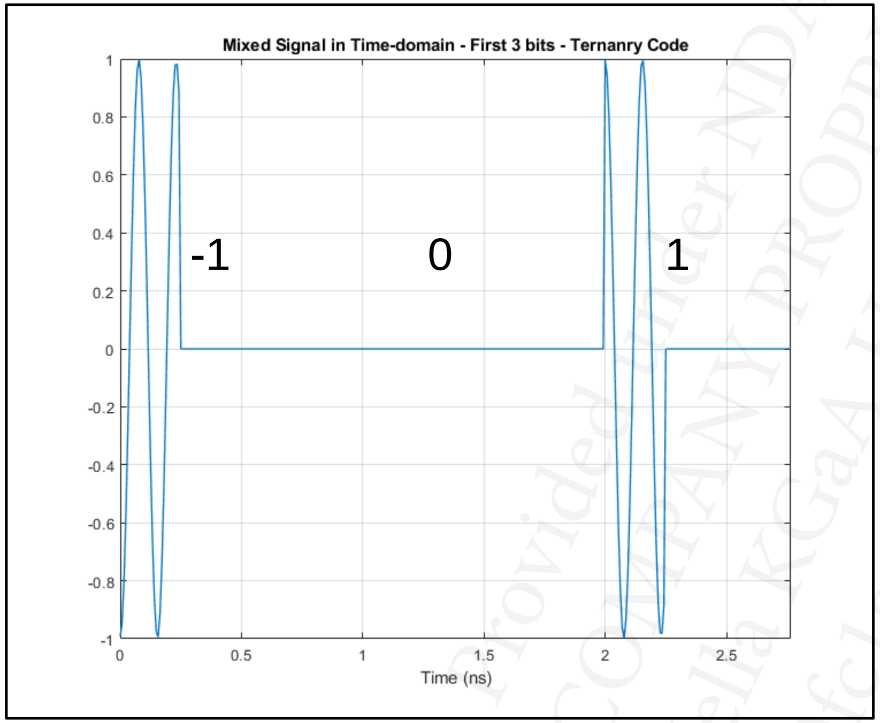




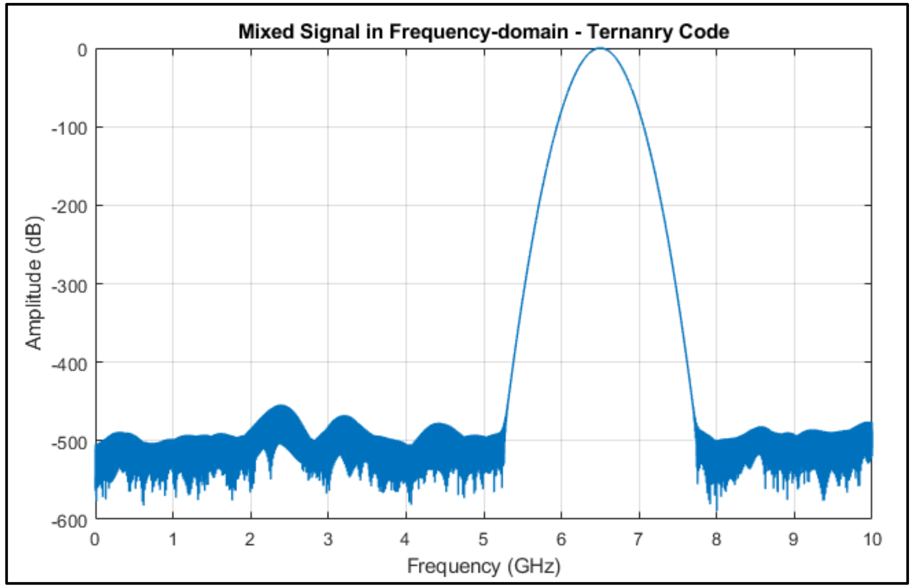
MIXED SIGNAL

- Mixing the carrier signal with the pulse modulates the signal to a higher frequency and allows transmission through space
- Radar pulse shape achieves near zero side-lobes contrary to a Ranging pulse





*Not actual mixed signal, shown for illustration of modulation only



Provided under NDA only
COMPANY PROPRIETARY
Hella KGaA Hueck & Co.
a0fc1fac-6480-49a3-b722-4b9e92ea31ae

IEEE Frame Format

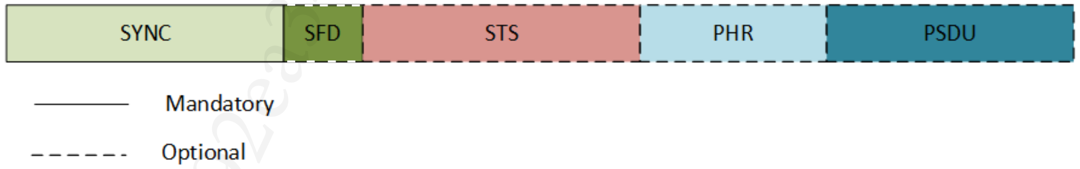
SECURE CONNECTIONS
FOR A SMARTER WORLD

Provided under NDA only
COMPANY PROPRIETARY
Hella KGaA Hueck & Co.
afc1ac-6480-49a3-8722-4b9e92ea34ae

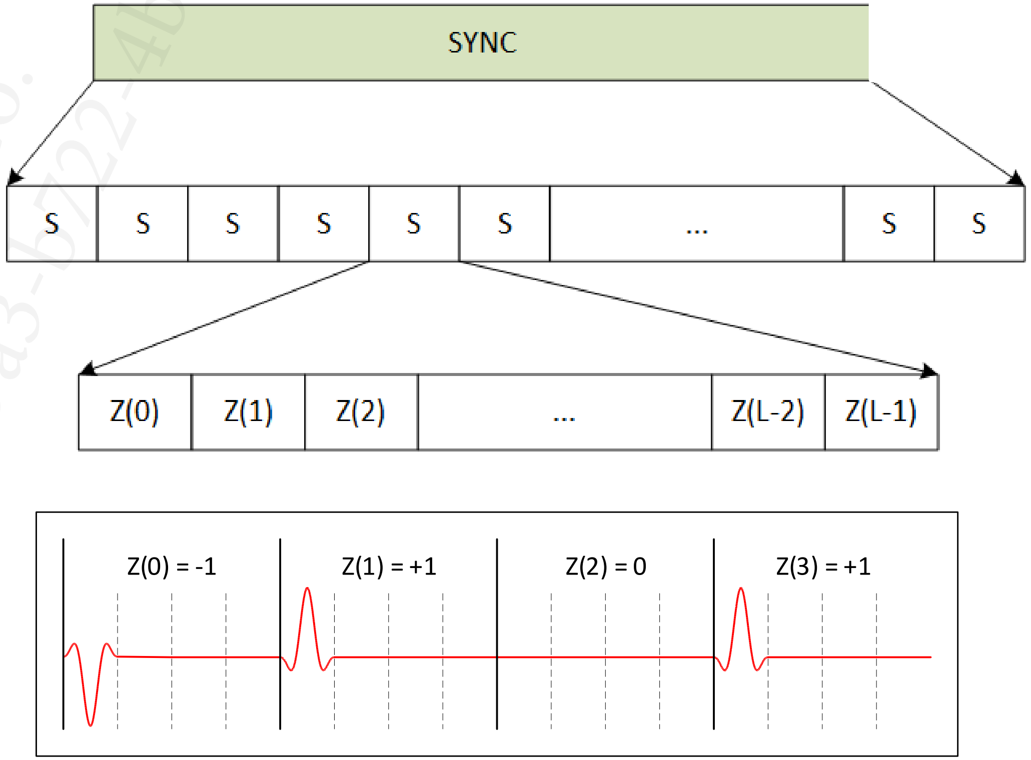
UWB RADAR FRAME

- Frames optimized for UWB Radar operation
  - No SFD, STS and no Payload
- SYNC:
  - 124.8 MHz PRF
- Pulses can be coded as +1, 0, -1 (Ternary Code)
- Or can be coded as +1, -1 (Binary Code)

IEEE Standard UWB frame format


SYNC
SFD
STS
PHR
PSDU
Mandatory
Optional

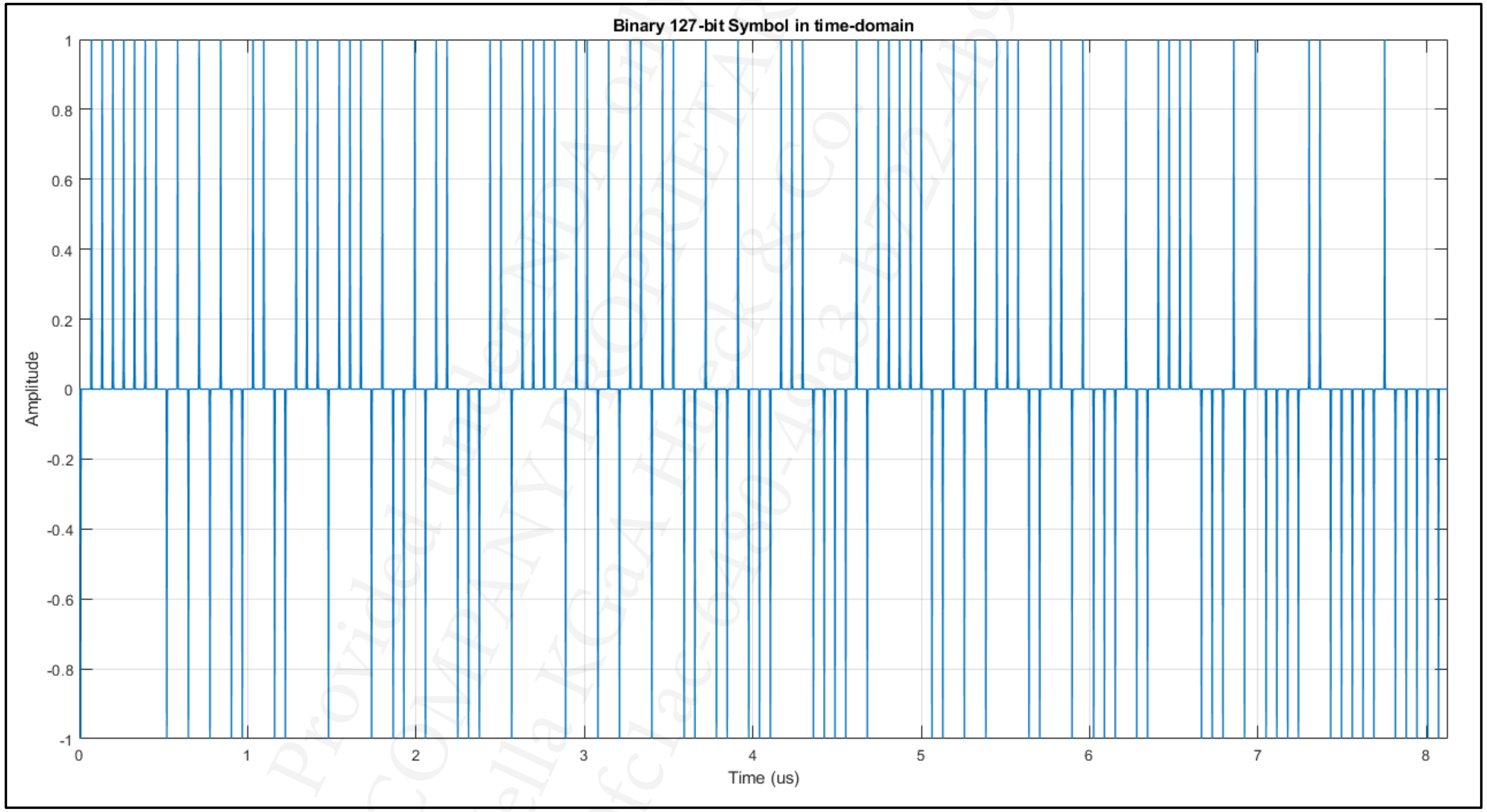
Radar Frame


SYNC
S
S
S
S
S
S
...
S
S
Z(0)
Z(1)
Z(2)
...
Z(L-2)
Z(L-1)
Z(0) = -1
Z(1) = +1
Z(2) = 0
Z(3) = +1

BINARY SYMBOL

- Exemplary Binary code with 127 bits:

[-+++++++-+-+-+--++--+++-+++-+--+-++---++-++++-++-+-++-++--+--+---+++----+-++++++--+-+-+++--++-+---+--++++---+-+----++-----+-----]

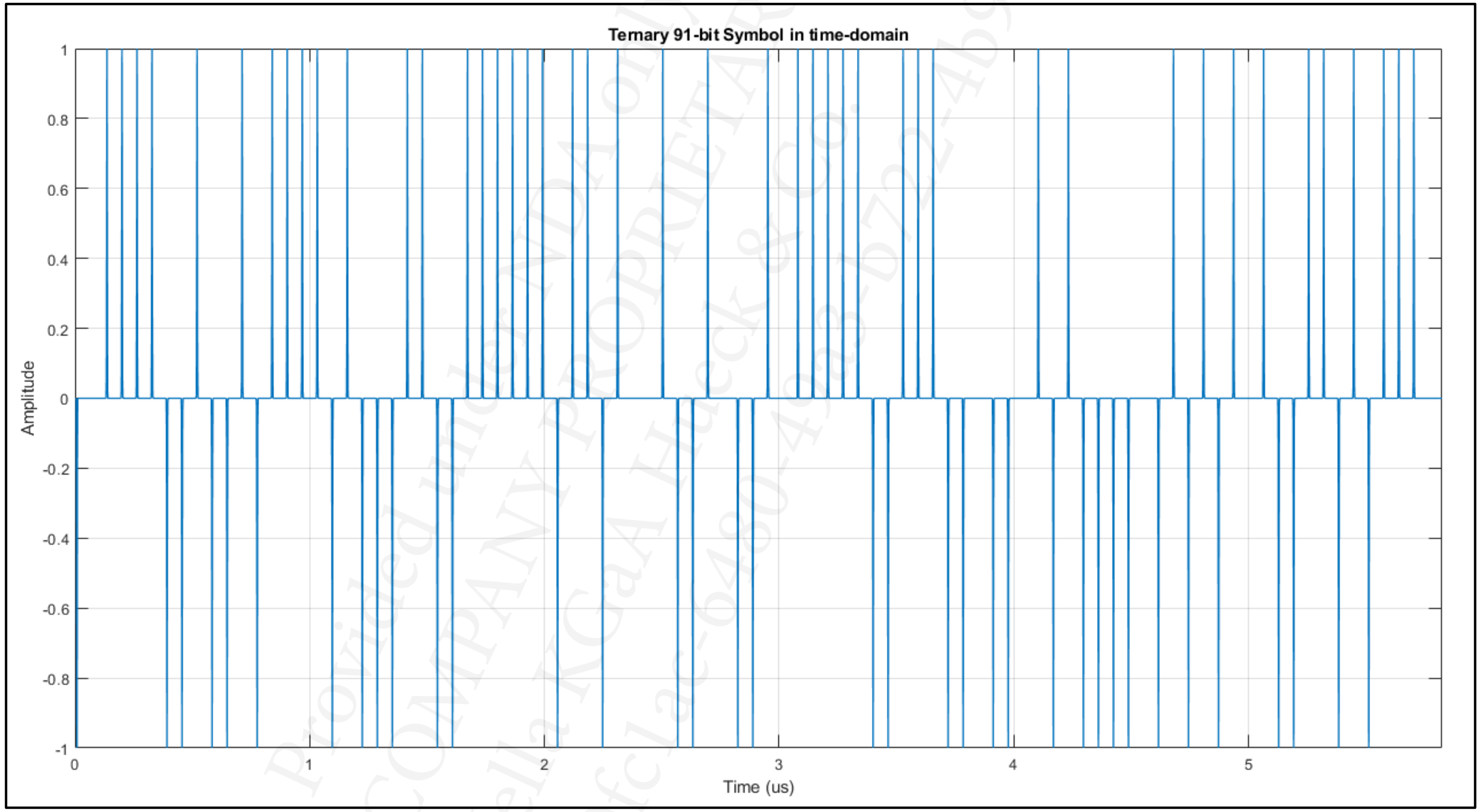


Provided under NDA only
COMPANY PROPRIETARY
Hella KGaA Hueck & Co.
eadfc1ac-6480-49a3-b722-4b9e92ea34ae

TERNARY SYMBOL

- Exemplary Ternary Code with 91 bits:

[-0++++--+--+-++++-+---++--++++++-++-+00+--+0--+0+++++--+++--0--0+-+----0-+-+-+0+--++-+-+++0]



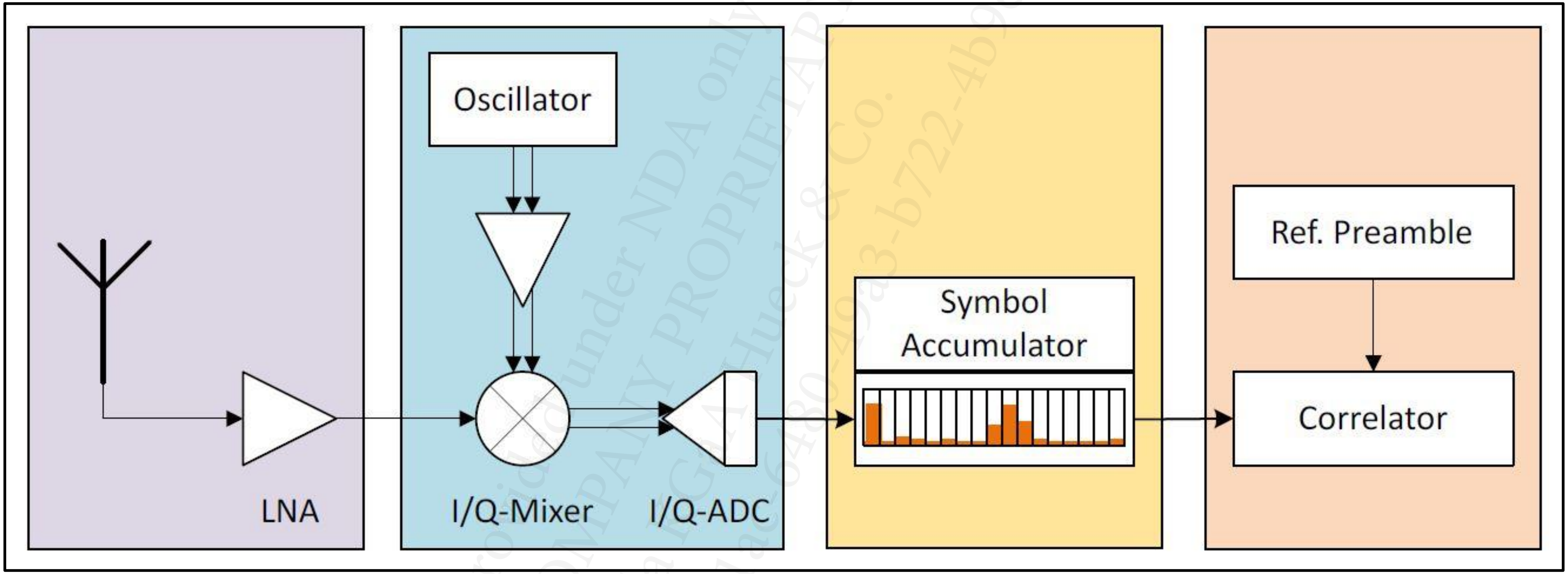
Provided under NDA only
COMPANY PROPRIETARY
Hella KGaA Hueck & Co.
ea0fc1ac-6480-4983-b722-4b9e9ea34ae

UWB Radar Receiver Concept

SECURE CONNECTIONS
FOR A SMARTER WORLD

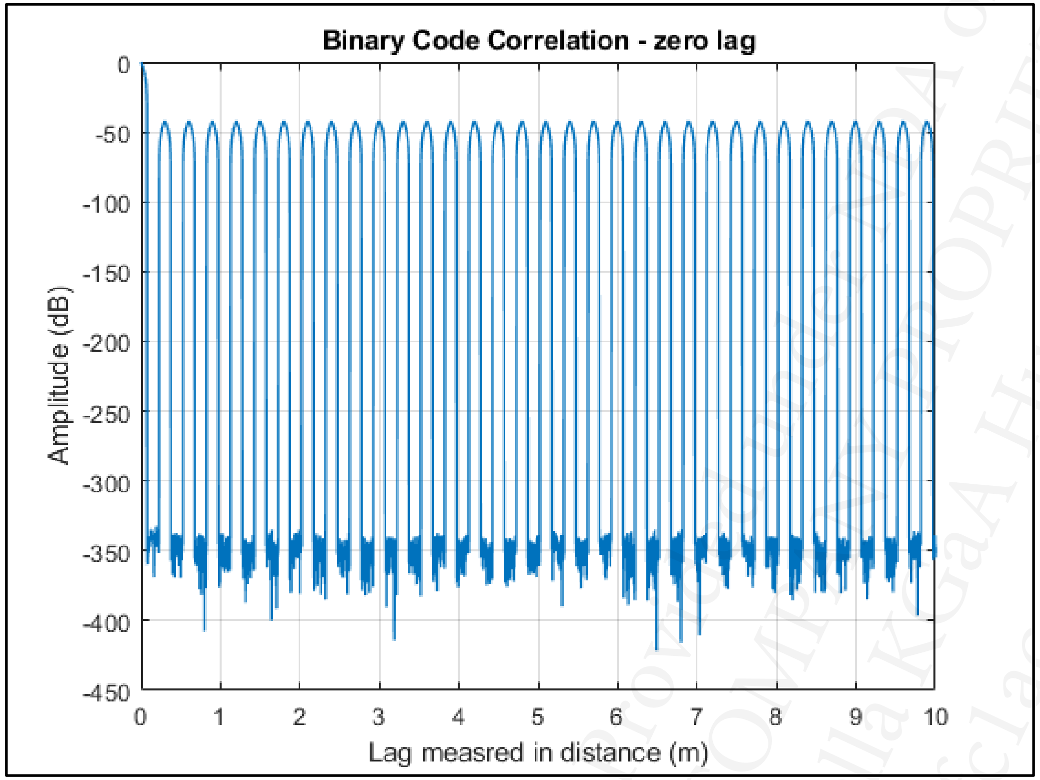
UWB RADAR RECEIVER

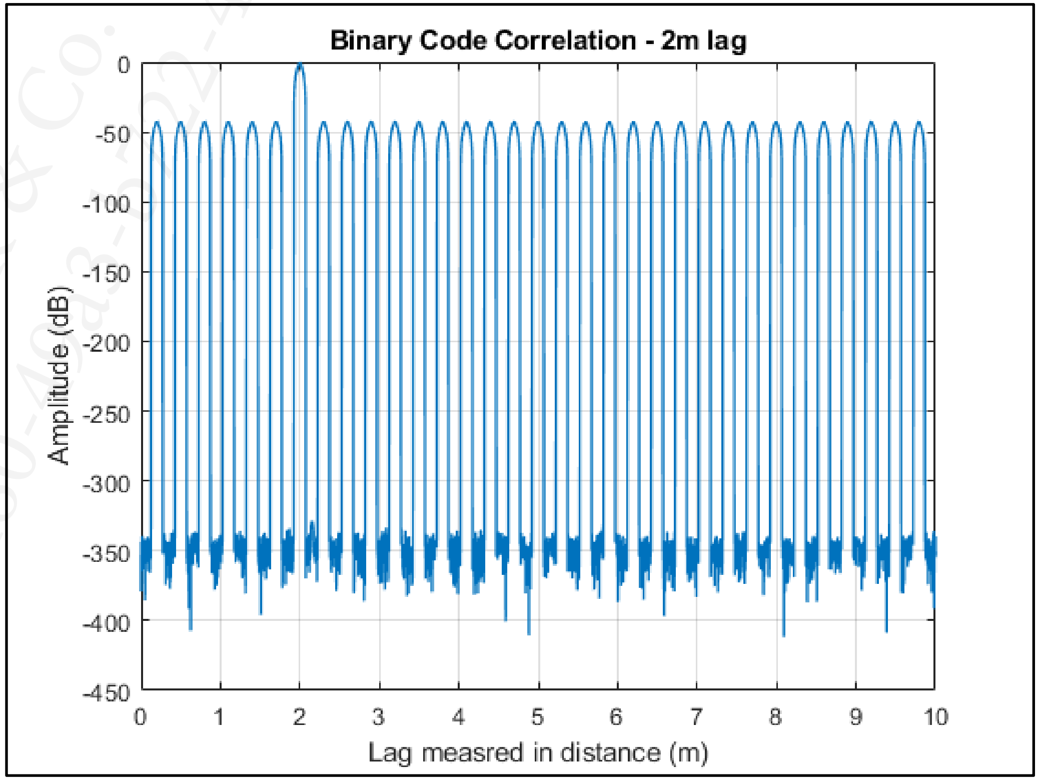
- UWB Radar receiver accumulates then correlates
- Software implementation of the correlator with higher quantization increases SNR


Oscillator
Ref. Preamble
Symbol Accumulator
Correlator
LNA
I/Q-Mixer
I/Q-ADC

CHANNEL IMPULSE RESPONSE – BINARY CODE

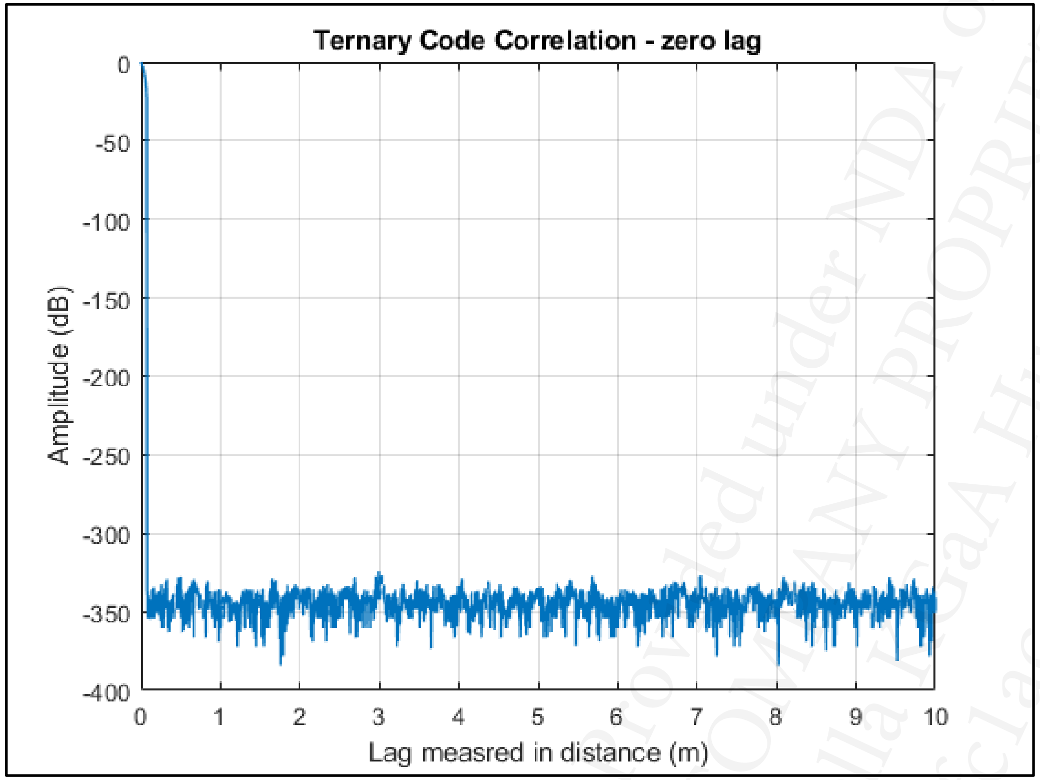
- Binary codes suffer from correlation artefacts that significantly increases the noise floor
- Targets need to be much-more reflective to be observed above the artefact level near ~ -50 dB
- A target at a given distance is a lag in the correlator input signal, hence shifting the correlation peak
- The influence of the pulse shape can be seen clearly here

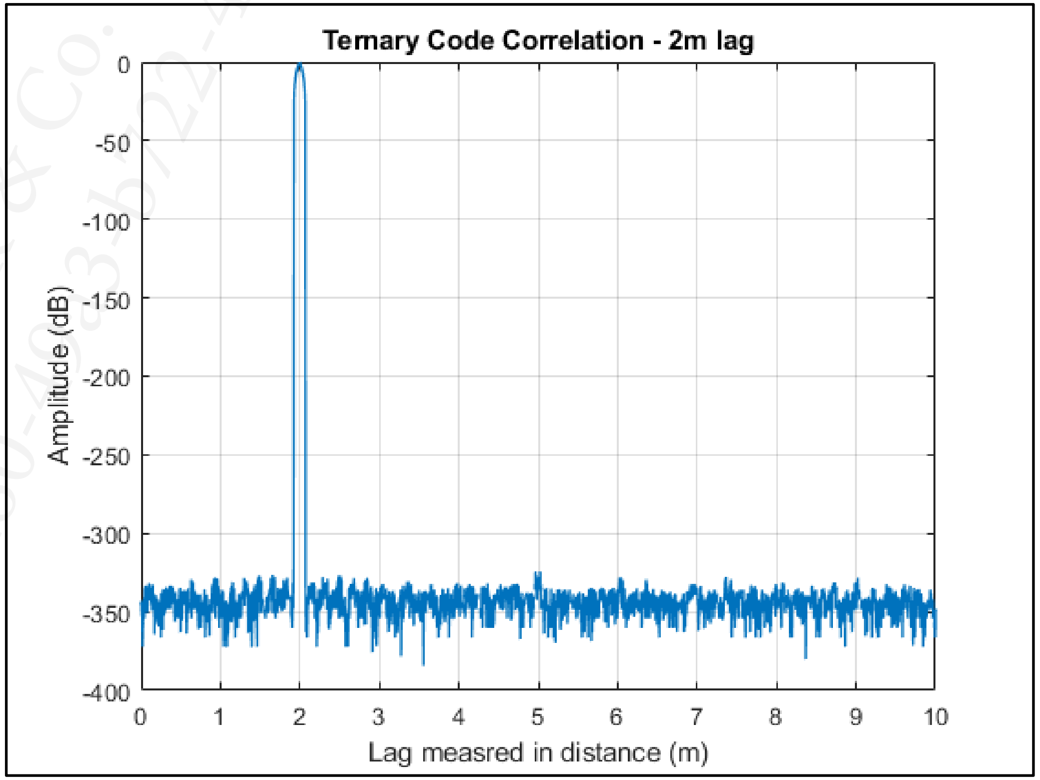




CHANNEL IMPULSE RESPONSE – TERNARY CODE

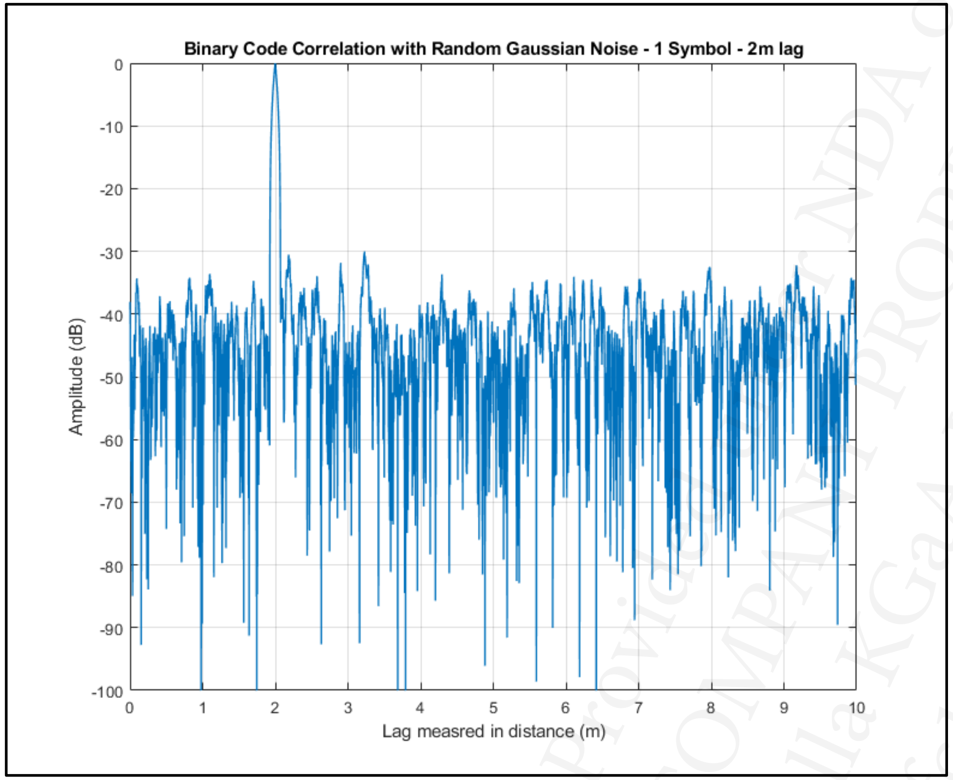
- Ternary Codes achieves near perfect correlation
- Allows for a much-reduced noise floor, hence detection of less reflective targets is possible

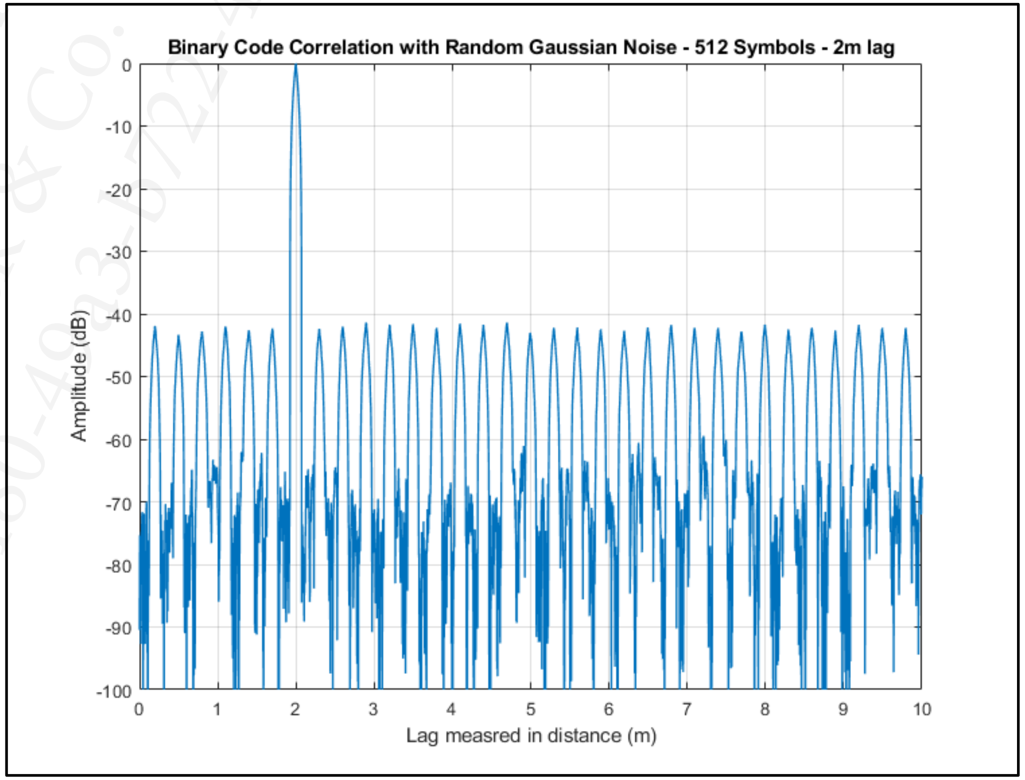




ACCUMULATOR EFFECT - BINARY

- Accumulation (coherent integration) over 512 symbols increases SNR by about ~ 10 dB
- Binary Codes correlation artefact remains

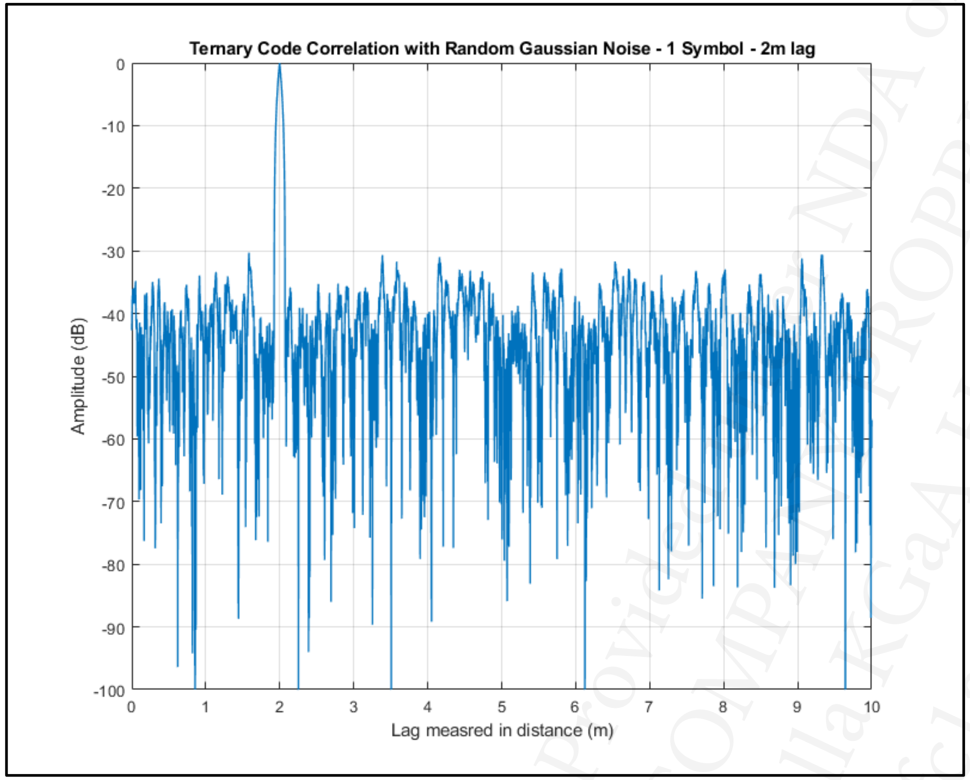


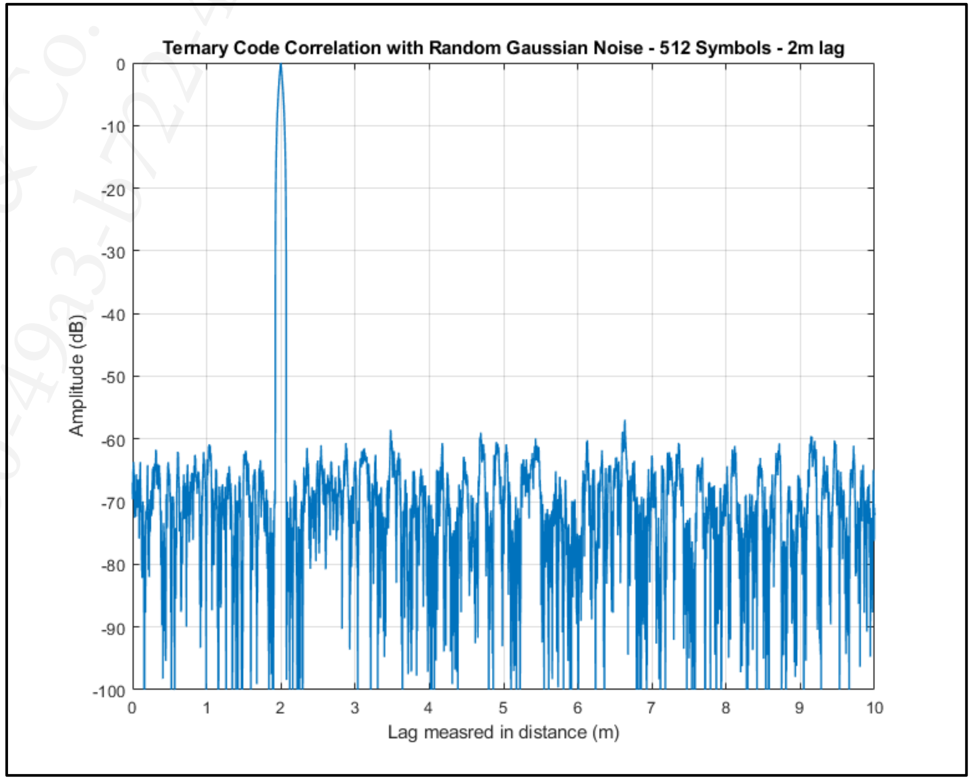


Provided under NDA only
COMPANY PROPRIETARY
Hella KGaA Hueck & Co.
a0fc1ac-6480-49a3-b722-4b9e92e34ae

ACCUMULATOR EFFECT – TERNARY

- Accumulation (coherent integration) of 512 symbols increases SNR by roughly ~ 10+20 dB
- No correlation artefact is present with ternary codes (~ 20 dB)





Provided under NDA only
COMPANY PROPRIETARY
Hella KGaA Hueck & Co.
a0fc1ac-6480-4923-b722-4b9e92e34ae

Signal Processing

SECURE CONNECTIONS
FOR A SMARTER WORLD

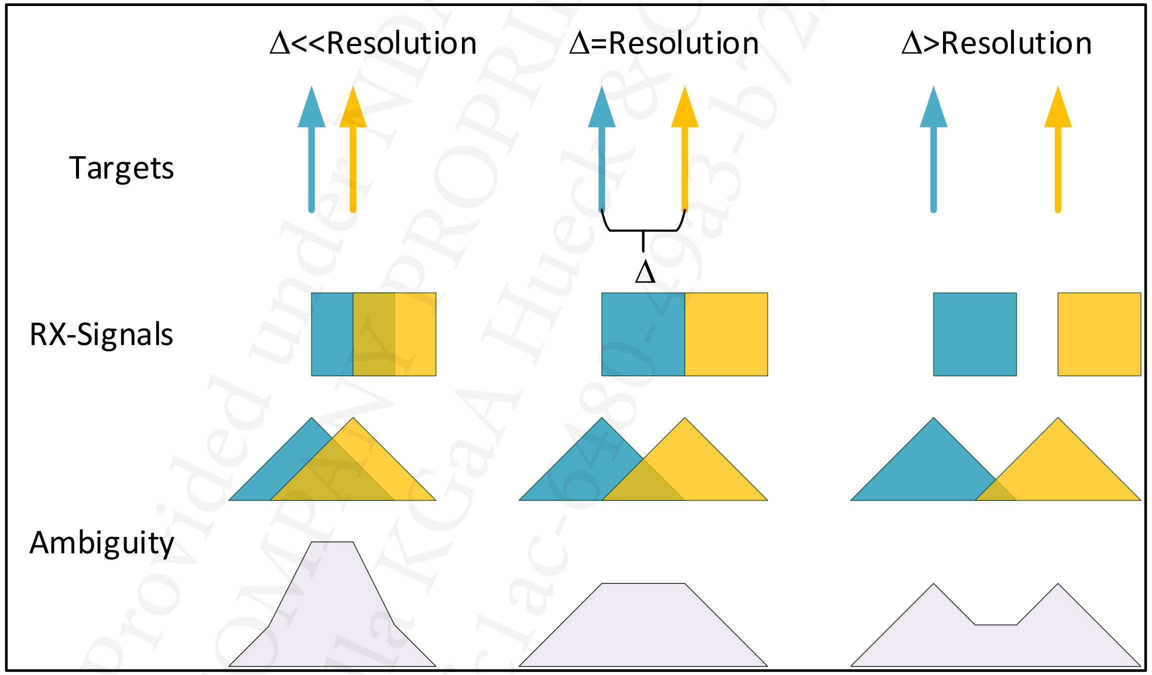
RANGE

- Maximum Range is related to the Symbol Repetition Frequency:

$R_{max} = \frac{c}{2 * SRF}$ -> well over 100 meters for 512 symbols for both binary and ternary codes

- Range resolution is related to the bandwidth of the transmitted pulse:

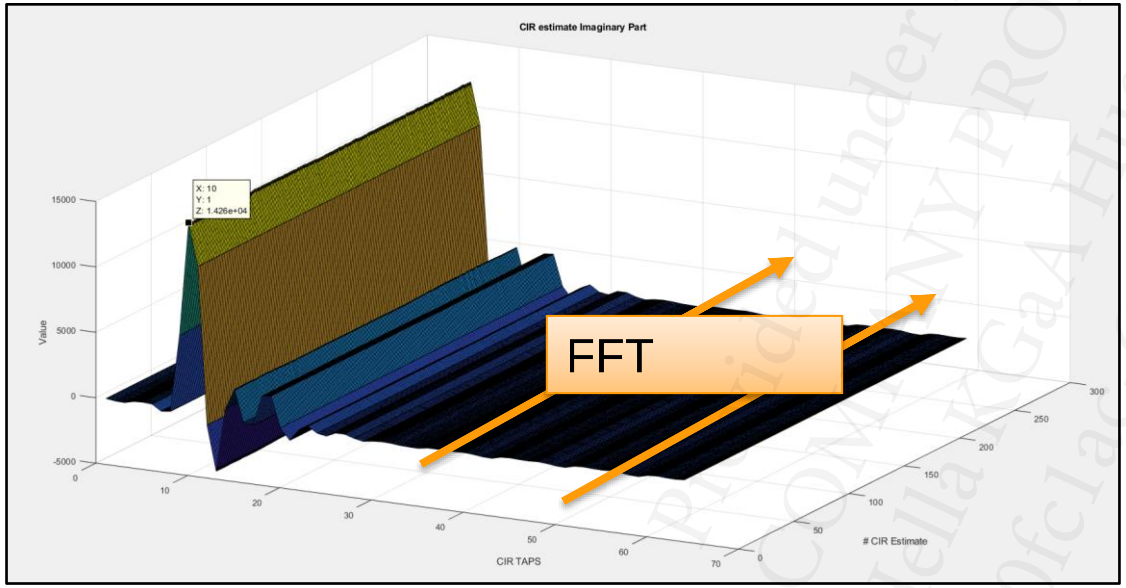
$R_{res} = \frac{c}{2 * B}$ = ~ 30 cm -> can be reduced by detecting the leading edge

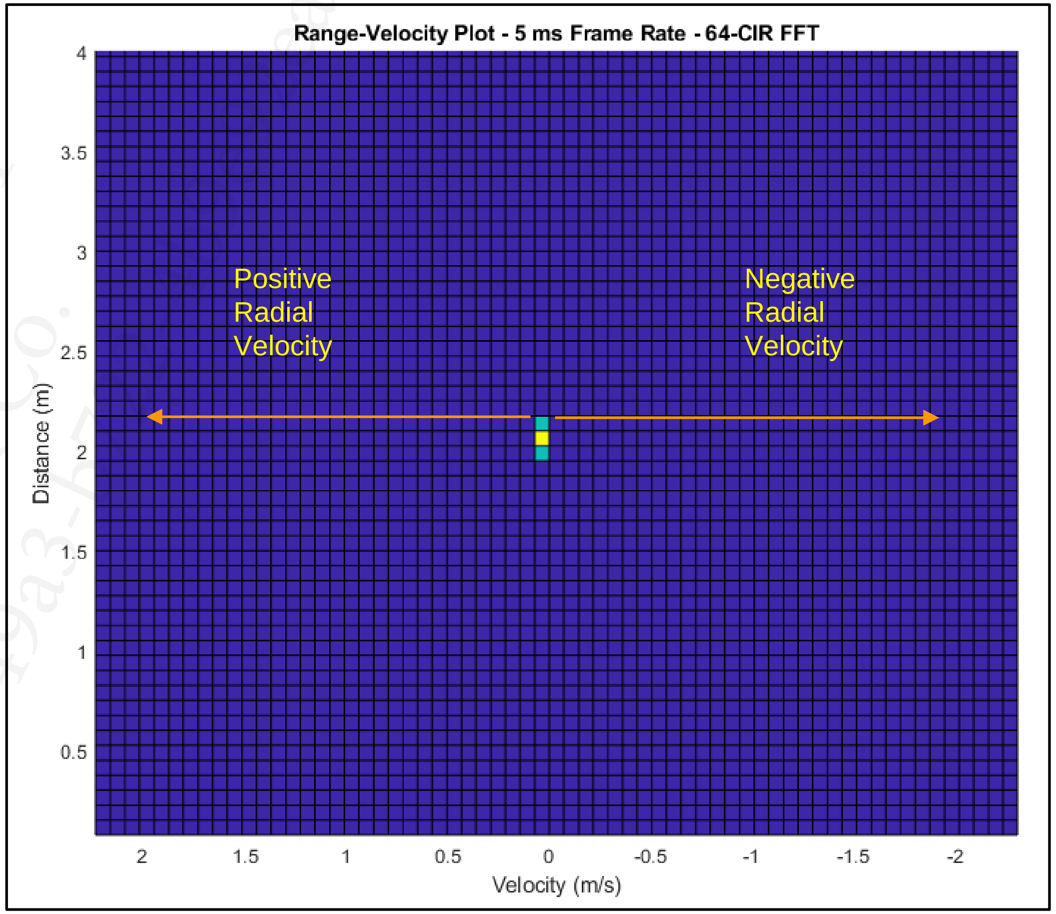

Δ<<Resolution
Δ=Resolution
Δ>Resolution
Targets
Δ
RX-Signals
Ambiguity

Provided under NDA only
COMPANY PROPRIETARY
Hella KGaA Hueck & Co.
a0fc1ac-6480-29a3-b722-b692ea34ae

VELOCITY - 1

- The Velocity can be resolved by applying an FFT across a stack of Channel Impulse Responses
- Maximum Velocity is related to the Frame Repetition Rate:
- $V_{max} = \frac{\lambda}{4 * FRR}$
- Velocity Resolution is related to the total observation time:
- $V_{res} = \frac{\lambda}{2 * T_f}$





Provided under NDA only
COMPANY PROPRIETARY
Hella KGaA Hueck & Co.
a0fc1ac6480-49a3-
34ae

VELOCITY - 2

- Total observation time:

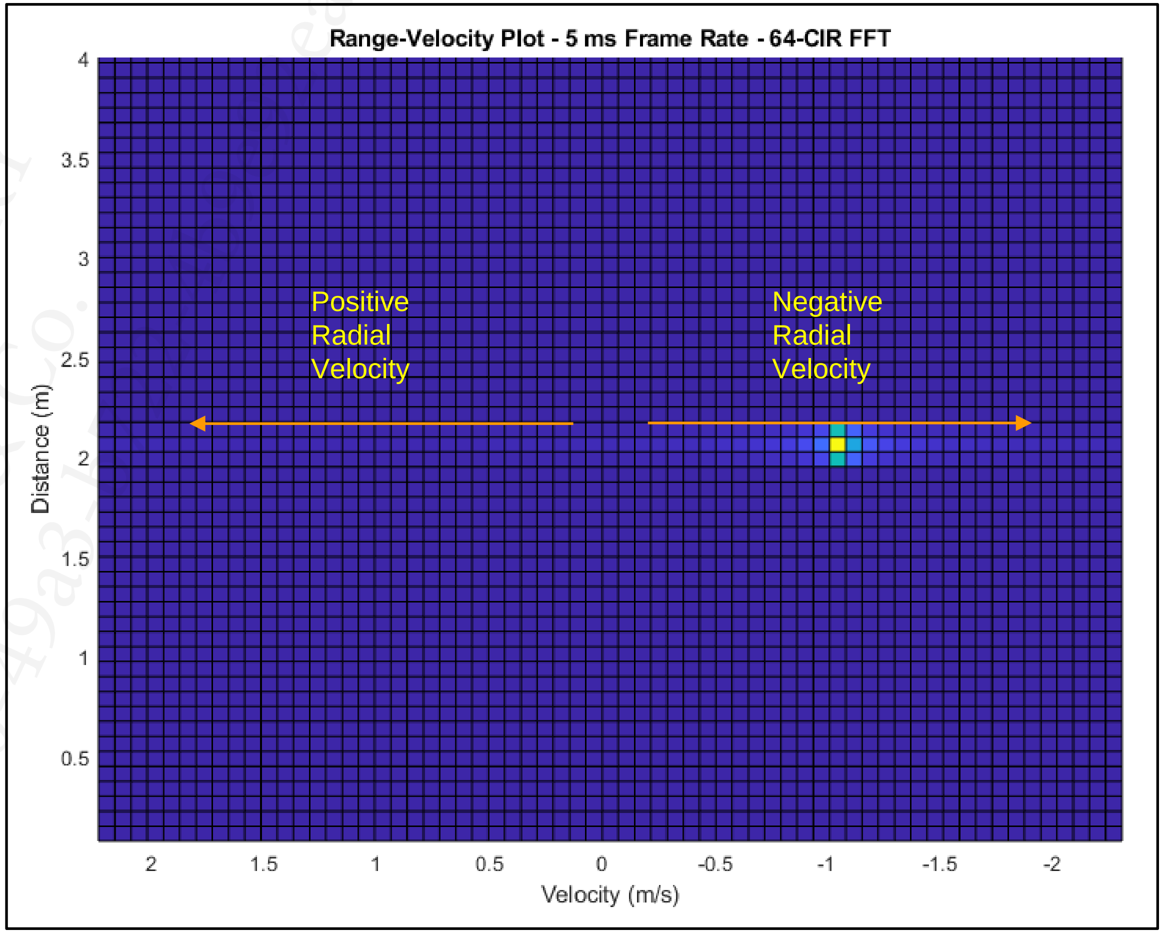
  $T_f = 5\ ms * 64 = 0.32\ s$

  Results in:

  $V_{max} = 2.3\ m/s$

  $V_{res} = 0.072\ m/s$

- Short observation time suitable for rapidly moving targets



Provided under NDA only
COMPANY PROPRIETARY
Hella KGaA Hueck & Co.
a0fc1ac-6480-49a3-...ea34ae

VELOCITY -3

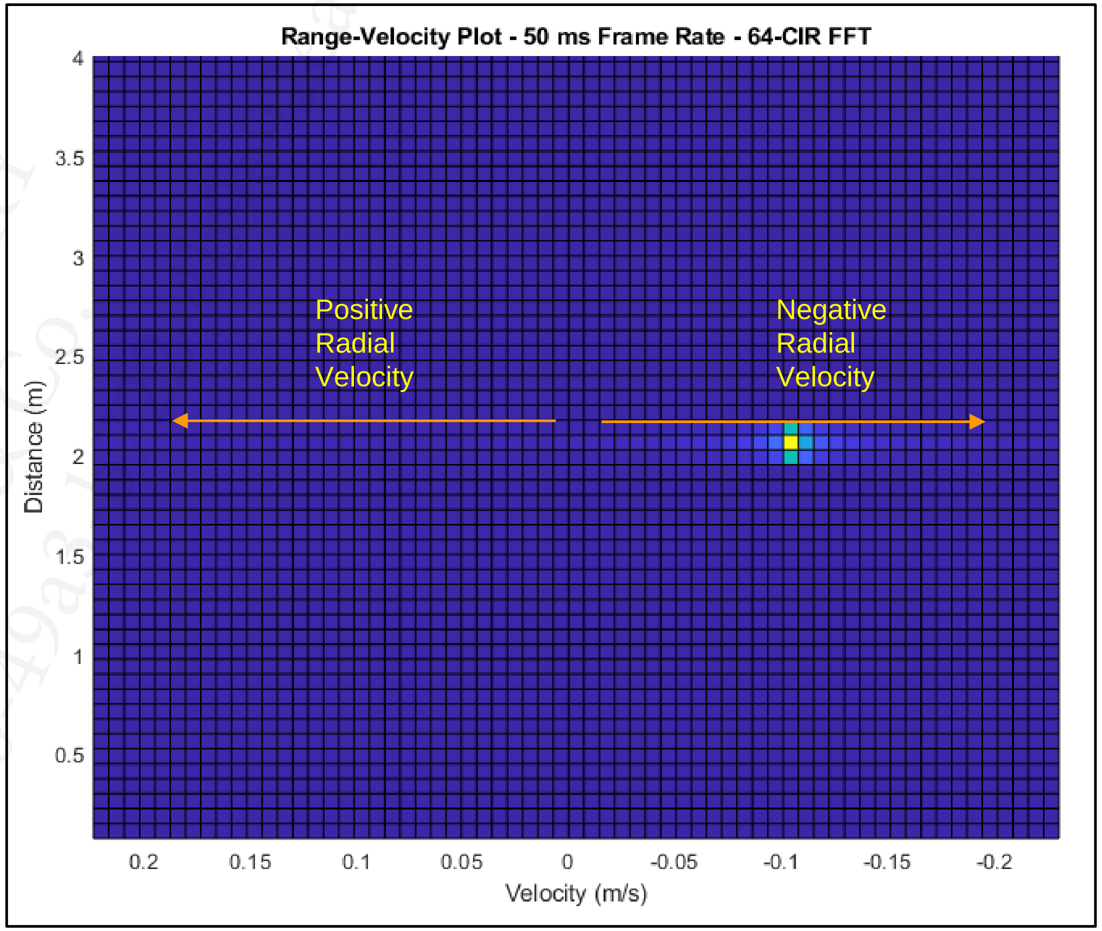
- Total observation time:

$T_f = 50\ ms * 64 = 3.2\ s$

Results in:

$V_{max} = 0.23\ m/s$

$V_{res} = 0.0072\ m/s$



Provided under NDA only
COMPANY PROPRIETARY
Hella KGaA Hueck & Co.
a0fc1ac-6480-49a3-...-34ae

VELOCITY - 4

- Total observation time:

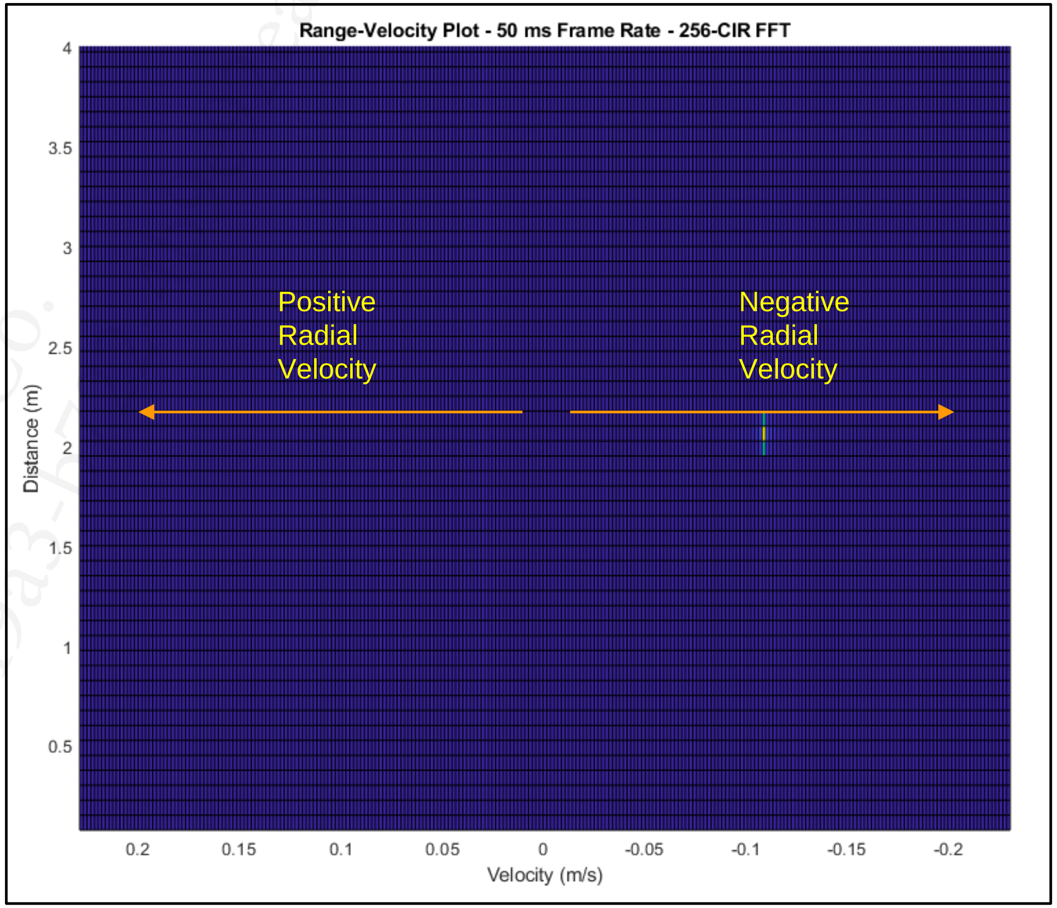
  $T_f = 50\ ms * 256 = 12.8\ s$

  Results in:

  $V_{max} = 0.23\ m/s$

  $V_{res} = 0.0018\ m/s$

- Long observation time for precise detection of slow-moving/changing targets



Provided under NDA only
COMPANY PROPRIETARY
Hella KGaA Hueck & Co.
a0fc1ac-6480-49a3-b5...a34ae

ANGLE OF ARRIVAL

- Increased path distance is translated as a phase shift in the Channel Impulse Response data
- The Phase Difference of Arrival (PDoA) can be calculated by subtracting the phases from each paths' CIR at a given distance
- The AoA can hence be calculated as:

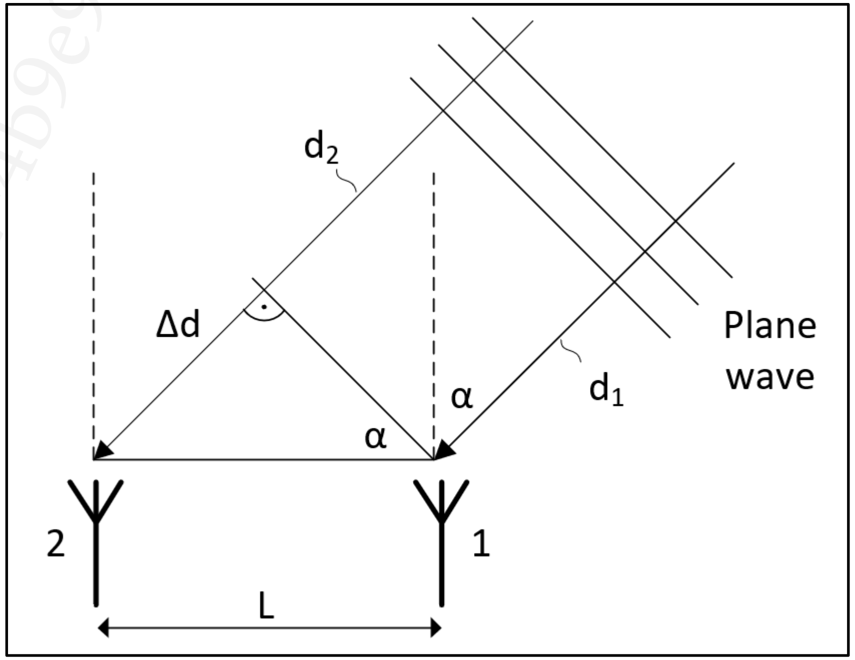
$$\alpha = \arcsin\left(\frac{\lambda\Delta\phi}{L2\pi}\right)$$

- With $L = \frac{\lambda}{2}$, the equation is reduced to:

$$\alpha = \arcsin\left(\frac{\Delta\phi}{\pi}\right)$$

- Example:

For target inducing a phase shift of 30º degrees, the AoA is about ~ 10º degrees from boresight.



Provided under NDA only
COMPANY PROPRIETARY
Hella KGaA Hueck & Co.
a0fc1ac-6080-49a3-b722-409e92ea34ae

Q&A

SECURE CONNECTIONS
FOR A SMARTER WORLD

Provided under NDA only
COMPANY PROPRIETARY
Hella GmbH & Co. KGaA Hueck & Co.
afc1ac-6480-49a3-b722-4b9e92ea34ae

RESTRICTED

DISCLAIMER

All information provided by NXP is accurate to the best of NXP's knowledge and will not operate to create or increase any NXP obligation. All information is provided "AS IS" and NXP makes no representation or warranty, express or implied, of accuracy, completeness, that products will be suitable for any specified use, or that the information, test results, analysis or assessments are reliable without further testing or modification by the customer. NXP will not be liable for any damage or loss arising from, in connection with or incident to any information or assistance provided by NXP. Customers are responsible for the design and operation of their applications and products and are responsible to provide appropriate design and operating safeguards to minimize risks associated with their applications and products

Unless otherwise recorded in a signed, written agreement, all sales transactions by NXP are exclusively subject to NXP's Terms and Conditions of Commercial Sale ("NXP Terms") published at: www.nxp.com/profile/terms/index.html. NXP explicitly rejects and disregards any terms and conditions of customer that add to, or differ from, NXP's Terms irrespective of when customer raises its terms.

Provided under NDA only
COMPANY PROPRIETARY
Hella KGaA Hueck & Co
a0fc1ac-6400-49a3-b72-48e2234ae

SECURE CONNECTIONS
FOR A SMARTER WORLD

Provided under NDA only
COMPANY PROPRIETARY
Hella KGaA Hueck & Co.
a0fc1ac-6480[illegible]3-b722-4b9e92ea34ae